

## MARKING SCHEME

### NETWORK DESIGN AND MANAGEMENT CAT 2

#### How the marks were allocated

- **One mark per point asked.** The number of marks equals the number of items the question asks for ("list four" = 4 marks, "state two" = 2 marks, "write the three commands" = 3 marks).
- **Each calculated item earns 1 mark.** Award the mark for the correct answer with the formula or working shown. If the answer is correct but no working is shown, award ½. If the working is correct but there is an arithmetic slip, award ½.
- **Each command earns 1 mark.** Each command is one action the device must perform.
- **Error carried forward:** a wrong answer used correctly in a later part is not penalised again.

**1. (4 marks)** 1 mark each: Application, Transport, Internet, Network Access (Link).

*Total = 4 layers asked, 1 mark each.*

**2. (2 marks)** Any two, 1 mark each: logical addressing, routing / path determination, packet forwarding, fragmentation and reassembly.

*Total = 2 functions asked, 1 mark each.*

**3. (2 marks)**

(i) Dividing IP addresses into fixed classes (A to E), each with a fixed default mask. (1)

(ii) Allocating addresses using a variable-length prefix (e.g. /26) instead of fixed classes. (1)

*Total = 2 definitions asked, 1 mark each.*

**4. (2 marks)** Any two differences, 1 mark each, each stating both sides:

- Static is assigned manually; dynamic is assigned automatically by a DHCP server.
- Static does not change; dynamic may change when the lease expires.

- Static suits servers, printers and routers; dynamic suits user PCs, laptops and phones.
- Static needs more administration; dynamic needs less.

*Total = 2 differences asked, 1 mark each.*

**5. (2 marks)** Any two differences, 1 mark each, each stating both sides:

- IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses.
- IPv4 is written in dotted decimal; IPv6 in hexadecimal separated by colons.
- IPv4 has about 4.3 billion addresses; IPv6 has a vastly larger address space.
- IPv4 uses broadcast; IPv6 does not.

*Total = 2 differences asked, 1 mark each.*

**6. (4 marks)**

(a) 1 mark each (3 marks): Class A **255.0.0.0**, Class B **255.255.0.0**, Class C **255.255.255.0**.

(b) **Class C** (1 mark), because the first octet 192 lies in the range 192 to 223.

*Total = 3 masks + 1 class = 4 items, 1 mark each.*

**7. (6 marks)**

(a) 1 mark each (3 marks):

- (i) Host bits  **$h = 32 - \text{prefix length}$**
- (ii) Total addresses =  **$2^h$**
- (iii) Usable hosts =  **$2^h - 2$**

(b) 1 mark each, for the working and the answer (3 marks):

- (i)  **$h = 32 - 26 = 6$**
- (ii)  **$2^6 = 64$**
- (iii)  **$64 - 2 = 62$**

*Total = 3 formulas + 3 calculations = 6 items, 1 mark each.*

**8. (4 marks)** 1 mark each. Working: the prefix gives the number of 1s, and the last octet value is the sum of the bit values.

- (a) /24 = **255.255.255.0**
- (b) /26 = 24 + 2 ones in the last octet ( $128 + 64 = 192$ ) = **255.255.255.192**
- (c) /28 = 24 + 4 ones ( $128 + 64 + 32 + 16 = 240$ ) = **255.255.255.240**
- (d) /30 = 24 + 6 ones ( $128 + 64 + 32 + 16 + 8 + 4 = 252$ ) = **255.255.255.252**

*Total = 4 conversions asked, 1 mark each.*

**9. (3 marks)** 1 mark each. Working: count the 1s in the binary form of the mask.

- (a)  $255.255.255.0 = 24$  ones = **/24**
- (b)  $255.255.255.224 = 24 + 3$  ones ( $128 + 64 + 32 = 224$ ) = **/27**
- (c)  $255.255.255.252 = 24 + 6$  ones ( $252 = 11111100$ ) = **/30**

*Total = 3 conversions asked, 1 mark each.*

**10. (4 marks)** 1 mark each:

- (a) Formula  $2^h - 2 \geq 50$ . For  $h = 5$ ,  $2^5 - 2 = 30$  (too small). For  $h = 6$ ,  $2^6 - 2 = 62$  (enough).  **$h = 6$**
- (b) Prefix =  $32 - h = 32 - 6 =$  **/26**
- (c) /26 = **255.255.255.192**
- (d) Usable hosts =  $2^6 - 2 = 64 - 2 =$  **62**

*Total = 4 values asked, 1 mark each. Each answer builds on the previous one.*

**11. (2 marks)** Any two, 1 mark each:

- It uses IP addresses efficiently, because the prefix length can match the number of hosts needed.
- It reduces the size of routing tables through route aggregation (summarisation).
- It is not limited to the fixed Class A, B and C sizes.

- It slows the exhaustion of IPv4 addresses.

*Total = 2 advantages asked, 1 mark each.*

**12. (2 marks)**

(a) **DHCP** (Dynamic Host Configuration Protocol). (1)

(b) Any one: a router interface must keep a fixed, predictable address because other devices use it as their default gateway; routing and management depend on it; a changing address would break connectivity. (1)

*Total = 2 items asked, 1 mark each.*

**13. (6 marks)** 1 mark each:

(a) Router(config)# hostname R1

(b) R1(config)# enable secret kcse2026

(c) R1(config)# interface gigabitethernet 0/0

(d) R1(config-if)# ip address 192.168.10.1 255.255.255.192 (mark awarded only for the correct mask)

(e) R1(config-if)# no shutdown

(f) R1# copy running-config startup-config (or write memory)

*Total = 6 tasks asked, 1 mark each.*

**14. (3 marks)** 1 mark each:

Switch(config)# interface vlan 1

Switch(config-if)# ip address 192.168.10.2 255.255.255.192

Switch(config-if)# no shutdown

*Total = 3 commands asked, 1 mark each.*

**15. (4 marks)** 1 mark each:

R1(config)# router rip

R1(config-router)# version 2

R1(config-router)# network 10.0.0.0

R1(config-router)# network 192.168.10.0

(no auto-summary is good practice and may be accepted, but it is not required for the four marks.)

*Total = 4 commands asked, 1 mark each.*

### **Marks per Question**

- Question 1: 4 marks
- Question 2: 2 marks
- Question 3: 2 marks
- Question 4: 2 marks
- Question 5: 2 marks
- Question 6: 4 marks
- Question 7: 6 marks
- Question 8: 4 marks
- Question 9: 3 marks
- Question 10: 4 marks
- Question 11: 2 marks
- Question 12: 2 marks
- Question 13: 6 marks
- Question 14: 3 marks
- Question 15: 4 marks
- **Total: 50 marks**

### **Topic Coverage**

- **2.4.1 Network models:** Q1, Q2 (6 marks)
- **2.4.2 IP addressing (classful, IPv4 and IPv6, classes, CIDR, static and dynamic):** Q3 to Q12 (31 marks)

- **2.4.3 Switch and router configuration (initial setup, interfaces and IP addresses, RIP):** Q13, Q14, Q15 (13 marks)

The paper is back to **50 marks**, and the time is now 1 hour 40 minutes to keep 2 minutes per mark. I added 15 marks of new questions (Q8 to Q12), all on CIDR or addressing with no subnetting:

- **Q8 and Q9:** converting between prefix lengths and dotted-decimal masks (7 marks).
- **Q10:** finding the prefix for a required number of hosts (4 marks).
- **Q11:** advantages of CIDR (2 marks).
- **Q12:** DHCP and static addressing (2 marks).